D02 · Demand Planning & Forecasting

# 01 · Statistical forecasting & model selection

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## 1 WHAT IT IS

Statistical forecasting selects and parameterizes a quantitative model per demand pattern — exponential smoothing for smooth series, Croston/SBA for intermittent, regression or ARIMAX where drivers and autocorrelation are real — instead of imposing one default method across the whole portfolio.

## 2 WHY IT MATTERS

Croston 1972 **(Croston, 1972 — peer-reviewed)** proved that standard exponential smoothing applied to intermittent demand is structurally biased: it updates the estimate only when demand occurs, inflating the forecast right after an event and starving it between events. A portfolio where 30–60% of SKUs are intermittent — typical in spares, MRO, and long-tail finished goods — carries a hidden positive bias the aggregate MAPE never reveals, because the smooth A-items dominate the weighted error. The cost is structural: safety stock sized against a biased mean **over-buffers the wrong SKUs by 15–30%** while leaving the smooth movers under-protected.

## 3 HOW IT IS DONE

1. **Classify every SKU by demand pattern before choosing any model.** Use ADI (average inter-demand interval) and $CV^2$ to split smooth / erratic / intermittent / lumpy — the Syntetos-Boylan cut sits near ADI 1.32 and $CV^2$ 0.49. Pattern, not ABC value, drives method.
2. **Assign the model to the pattern, not the planner's habit.** Smooth -> Holt-Winters; intermittent and lumpy -> Croston or SBA; erratic with known drivers -> regression/ARIMAX. A default applied uniformly is a bias applied uniformly.
3. **Select on out-of-sample error, not in-sample fit.** Hold out the last N periods and run a rolling-origin backtest. In-sample MAPE rewards overfitting and lies about live accuracy.
4. **Choose the error metric per pattern.** MAPE is undefined at zero demand — worthless for intermittent series. Use MASE or RMSSE so the metric does not silently discard the SKUs that carry the bias.
5. **Re-fit on a cadence and re-classify on a slower one.** Refit parameters monthly; re-run the pattern classification quarterly. Demand patterns migrate as products age — a smooth mover becomes intermittent on its way to end-of-life.
6. **Govern the override as a forecast input.** Log every planner override, attribute it, and measure FVA (forecast value-add) against the statistical baseline so judgment that destroys accuracy becomes visible instead of invisible.

**WORKED EXAMPLE — ILLUSTRATIVE**

A distributor with **12,000 SKUs** ran single-method Holt-Winters across the catalog — **baseline: aggregate MAPE 18%, $34M inventory, 94% fill rate**. Reclassification found **41% of SKUs intermittent (4,920 SKUs)**, moved to SBA, with erratic movers shifted to ARIMAX. Results: **intermittent-class bias -22% -> -3%, safety stock on that class $8.1M -> $6.2M (-23%), fill rate 94% -> 96%** — while aggregate MAPE barely moved (**18% -> 17%**), proving the aggregate number was hiding the money the whole time.

**IN PRACTICE**

### Rolling-origin backtest as the selection gate

No model enters production without beating the naive seasonal benchmark on a rolling-origin backtest across the relevant horizon. Tournament the candidates per pattern class and let out-of-sample error pick — not the planner's preference, not the vendor's default.

### FVA governance of overrides

Track forecast value-add: compare the final forecast to the statistical baseline and to the naive forecast. Overrides that worsen accuracy are common and expensive; FVA makes them visible and lets you switch off the steps that destroy value.

### Re-classification cadence

Re-run pattern classification quarterly. Products migrate across patterns over their life; a model frozen at launch is forecasting a SKU that no longer exists. Tie re-classification to the lifecycle stage in your PLM/master data.

### Hierarchical reconciliation

Forecast at the level where the signal is statistically stable, then reconcile up and down the hierarchy. An intermittent SKU may be unforecastable alone but stable as part of a family — forecast the family, disaggregate with a coherent reconciliation method.

#### WHAT YOU WOULD USE

**STATISTICAL FORECASTING ENGINES**

**SAP IBP / o9 — enterprise tier-1** — Full statistical libraries with pattern-aware model selection, best-fit tournaments, and FVA reporting native. **(SAP, o9 — vendor)** The platform choice for portfolios where forecasting feeds an integrated S&OP/IBP plan.

**Blue Yonder Luminate Demand / Kinaxis** — Strong demand engines with ML-assisted model selection and segmentation. **(Blue Yonder, Kinaxis — vendor)** Tier-2 by footprint, not by capability — frequently the better fit for a single-region demand process.

**Forecast Pro / specialized statistical tools** — Purpose-built statistical forecasting with deep intermittent-demand methods (Croston, SBA, TSB). **(Forecast Pro — vendor)** The vertical choice when statistical rigor matters more than end-to-end planning integration.

**For operations that don't need a platform** — A documented model-selection procedure plus an open-source library — Python statsforecast (AutoETS, Croston, TSB) or R fable — run on a rolling-origin backtest. Standard tool, disciplined procedure: beats a misconfigured platform every time.

## 4 SO WHAT

**Operations that match the model to the demand pattern surface the bias the aggregate MAPE hides; operations that run one default method ship that bias straight into safety stock.** The aggregate accuracy number is the most expensive comfort in planning — it stays flat while the intermittent half of the portfolio quietly over- and under-buffers. Choosing the model per pattern is not statistical hygiene; it is where the working capital is.

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